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Combining ChatGPT and knowledge graph for explainable machine learning-driven design: a case study

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ABSTRACT

Machine learning has been widely used in design activities, enabling more informed decision-making. However, high-performance machine learning models, often referred to as 'black-box', result in a lack of explainability regarding predictions. The absence of explainability erodes the trust between designers and these models and hinders human-machine collaboration for desirable design decisions. Explainable AI focuses on creating explanations that are accessible and comprehensible to stakeholders, thereby improving explainability. A recent advancement in the field of explainable AI involves leveraging domain-specific knowledge via knowledge graph. Additionally, the advent of large language models like ChatGPT, acclaimed for their ability to output domain knowledge, perform complex language processing, and support seamless end-user interaction, has the potential to expand the horizons of explainable AI. Inspired by these developments, we propose the novel hybrid method that synergizes ChatGPT and knowledge graph to augment post-hoc explainability in design context. The outcome is the generation of more contextual and meaningful explanations, with the added possibility of further interaction to uncover deeper insights. The effectiveness of the proposed method is illustrated through a case study on customer segmentation.

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1. Introduction

Machine learning (ML) utilises vast data resources and advanced computational capabilities to discern patterns within data. Considering that design is an information-rich process generating significant data, many researchers have employed ML to simplify complex design tasks, enhance decision-making, and reveal latent insights (Li et al. 2018; Arjomandi Rad, Cenanovic, and Salomonsson 2023; Xie et al. 2023; Renzi, Leali, and DI Angelo 2017; Ali et al. 2023; Zhang et al. 2023; Zhou, Zhang, and Yu 2023). However, despite the considerable benefits associated with ML-driven design, several challenges persist, particularly concerning high-performance ML models that often act as black-box. These models, such as deep

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neural networks, can provide impressive results but are difficult to interpret the inner mechanism and explain the prediction due to their complex, multi-layered architecture, and non-linear decision-making processes (Adadi and Berrada 2018). Three primary concerns arise from this black-box nature:

- Data explainability: Design data – often heterogeneous and varying in quality, structure, and size – is collected from diverse channels with differing credibility levels. The obscurity of black-box models complicates the discernment of high-quality datasets, making AI training susceptible to biases, errors, and data corruption inherent in sub-datasets.
- Model explainability: The lack of transparency hinders understanding the model's underlying mechanisms, thereby eroding developers' intuition in model calibration. It becomes arduous to determine the sufficiency of training data for specific design challenges, recognise opportunities for model enhancement, or verify if the model has accurately discerned anticipated patterns (Ali et al. 2023). This opacity can compromise the synergy between data operations and design operations, potentially culminating in subpar or even adverse design outcomes.
- Post-hoc explainability: Designers, when leveraging data insights for their decisions, grapple with trusting black-box model outputs. The absence of post-hoc explainability imposes extra cognitive strain and influences trust, compelling designers to lean on their domain expertise or seek alternative validation methods, especially when certain AI predictions collide with the ground truth, common knowledge, and designer expectations.

Enhancing explainability is essential for fostering effective collaboration between humans and machines in design. Full automation in design is often not the ideal approach, not just because of the critical nature of the outcomes, but also because designers possess insights that can complement ML models output, ensuring optimal design decisions. The ML-driven design manifests human designers and AI collaborate to produce desirable design decisions. The efficacy of the ML-driven method is significantly influenced by the model performance and post-hoc explainability (Zhang, Liao, and Bellamy 2020). The ML model's performance, especially its accuracy and reliability in producing desired outputs, can either bolster or hinder design decisions. During the iterative training process, accuracy and reliability can be influenced by both data explainability and model explainability. These factors allow model developers to better understand model behaviour and the datasets employed during training.

Moreover, the post-hoc explainability – specifically, the extent to which a designer can understand the model output – plays a pivotal role in calibrating designer trust in design context. Knowing when to trust or question the AI output enables designers to judiciously apply their expertise, enhancing decision outcomes, especially in situations where the model performance might be poor. While explainable ML-driven design emphasises the importance of data, model, and post-hoc explainability, this research narrows its focus to post-hoc explainability. This emphasis stems from its profound impact on end users. Their ability to interpret ML model's outputs is paramount, making it essential to delve deeper into mechanisms that enhance post-hoc explainability.

Considering the imperative to bolster post-hoc explainability within the design, several critical challenges emerge. Primary among these is the imperative to develop explanations

that are contextually grounded. This entails crafting explanations tailored to the specific domain and the model in use, ensuring that they shed light on the model's reasoning while integrating pertinent background knowledge or information. Moreover, the explanation mechanism must be dynamic and interactive, offering designers the ability to receive on-the-spot support and pursue inquiries as needed. Furthermore, there is a notable scarcity of domain-specific knowledge in engineering design, and the tasks of accumulating and updating this knowledge demand considerable effort and are time-intensive. To address these challenges, this research proposes the integration of knowledge graph (KG) and ChatGPT. KG, structured graph-based representations, encodes knowledge through entities and their interrelationships, depicted as nodes and edges within the graph (Abu-Salih 2021). It has been proposed as an explainable AI (XAI) tool capable of generating knowledge-based explanations. These serve as background knowledge, enabling the AI system to better explain its decisions to users (Tiddi and Schlobach 2022). Concurrently, ChatGPT is recognised for its adeptness in contextual understanding and fluid interaction. It holistically processes conversational prompts, enabling contextually rich dialogues and positioning itself as a formidable chatbot tool (Wang et al. 2023). Besides, it is also employed in the generation of structured knowledge, which is promising to serve as a cornerstone for developing KG (Hu, Li, et al. 2023; Wang et al. 2023; Ling et al. 2023).

Against this background, our approach integrates KG with ChatGPT, aiming to furnish designers with more contextual-rich and interactive explanations on an instance level. This integration aspires to bridge the gap between AI outputs and human comprehension, enhancing the clarity and depth of AI-assisted design decisions. First, feature-based explanations derived from black-box model insights are integrated with domain-specific knowledge and analogous examples sourced from the KG. This amalgamation is orchestrated by crafting targeted prompts utilising ChatGPT, ensuring that the explanations provided are tailored to the domain and model-specific nuances, thus allowing for customised explanatory output. Second, ChatGPT is employed to facilitate the generation of structured domain knowledge and the construction of domain-specific KG. Third, the advanced conversational and linguistic capabilities of ChatGPT are leveraged to enhance the interactivity of the explanation process. As for the structure of this article: Section 2 delves into related research; Section 3 details the proposed framework; Section 4 presents a case study; Section 5 discusses the contributions and limitations of this study, and Section 6 concludes the study.

2. Related research

2.1. Post-hoc explainability in design

Post-hoc explainability refers to methods utilised to explain the ML models predictions (Adadi and Berrada 2018). The necessity and requirements for post-hoc explanations vary across different modes in which ML models are being used: emulating and discovering (Lai, Liu, and Tan 2020). In the emulating mode, ML models are crafted to mirror human intelligence. The primary utility of post-hoc explanations is to debug potential issues related to model robustness and data biases. Conversely, in the discovering mode, ML models are deployed to tackle tasks that are inherently complex or counterintuitive for humans. In these instances, there is a heightened need for clear and comprehensible explanations of

the model outputs to assist decision-making. It is crucial to recognise that these two modes – emulating and discovering – are not mutually exclusive but rather represent two ends of a spectrum. For instance, while an ML model might be emulating experts, the insights it provides could be a discovering mode for novices. When deploying ML models anywhere along this spectrum, it becomes crucial to provide explanations that are both contextual and interactive, tailored to meet the specific needs of the audience (Miller 2019).

In the study (Hu, Liu, et al., 2023), a novel framework was introduced, integrating XAI into the domain of customer segmentation – a design task in product planning. This approach enhanced both data and model explainability by using feature-based and data-based XAI, resulting in a positive impact on AI performance. The data-based XAI is employed to gain a deeper understanding of the datasets utilised during the training and design of AI models (Pradhan et al. 2022). The feature-based XAI quantifies the contribution of input feature values to the predictions made by the model, both on a global and instance level. Following the utilisation of feature-based XAI, another study introduced a pioneering customer-emotion-based approach aimed at evaluating service quality. This method leveraged feature-based XAI to elucidate the nuanced dynamic attributes of consumer emotions that directly influence service quality metrics (Guo et al. 2024). However, these research falls short in addressing post-hoc explainability. The provided explanations are predominantly static, focusing mainly on approximating the complex decision-making functions of the models. While this provides a fundamental grasp of the model's functionalities, it tends to overlook the pivotal aspects of context and background knowledge, which are crucial, especially during the early stages of design. In these initial phases, design problems are often ill-defined, and designers find themselves navigating through complex tasks that demand a comprehensive understanding beyond the model's representations of decisions. This is where the disparity between the model domain and the design problem domain becomes conspicuous. The model domain is characterised by the ML-driven processes that guide the AI's decision-making, often resulting in outputs that are abstract and detached from the real-world context of the design problem. On the other hand, the design problem domain is rich with contextual nuances, undefined parameters, and a need for experiential knowledge, elements that are crucial for making informed design decisions but are often missing from the model's considerations.

Interactive and contextual post-hoc explanations hold great potential in bridging the gap between the model domain and the design problem domain. By infusing the model's outputs with essential domain-specific knowledge, they render the information more pertinent and comprehensible for designers. Furthermore, these explanations transform designers from passive recipients to active participants in the explanatory process, fostering a dialogue with the AI system. This engagement is vital for aligning the AI's output with the designer's cognitive processes, ensuring that the explanations are not just accurate, but also rich in context and domain-specific knowledge. KG plays a crucial role in this context, providing a structured form of domain background knowledge that can be integrated into XAI systems to enhance the quality and relevance of explanations (Naqvi et al. 2024). Concurrently, technologies like ChatGPT, with their advanced language processing capabilities and contextual awareness (Wang et al. 2023), are instrumental in facilitating interactive explanations and providing additional context where necessary. To maximise the benefits of post-hoc explainability in design, it is essential to leverage the combined strengths of KG and ChatGPT.

2.2. Knowledge graph

KG is designed to encapsulate and structure complex information about entities and their interrelationships. Modelled as graph structures, they comprise nodes representing entities and edges denoting relationships, often structured in the form of triples (Abu-Salih 2021). The development of design-specific KG goes through two stages: design knowledge acquisition and KG construction. In design knowledge acquisition, design knowledge can be acquired from three sources, namely design documents, user-generated content, and generic KG (Liu et al. 2022). KG construction is a two-step process. Initially, a domain ontology is constructed, which serves as the structural blueprint for the knowledge representation. Subsequently, knowledge data is processed into factual triples, typically following the (subject, predicate, object) or (head, relation, tail) format, to augment this ontology (Zhu et al. 2022). As new data continues to emerge, the KG evolves through the expansion of the established domain ontology.

Researchers propose incorporating KG at different stages of the explainability process, namely, before modelling for datasets, during modelling for models, and after modelling for post-hoc explanations (Rajabi and Etminani 2022). Considering the complexities involved in integrating large-scale knowledge into black-box models, these models are often characterised by their integration of post-embedded knowledge (Tiddi and Schlobach 2022). Nevertheless, relying solely on KG for generating model-specific explanations presents challenges, as KG primarily contains external knowledge and may not fully align with a model's internal mechanism at an instance level. This discrepancy can lead to explanations that lack fidelity. To address this, this paper proposes a knowledge fusion approach by combining feature-based XAI for extracting model-specific insights with KG.

2.3. The combination of knowledge graph and large language models

Gathering design knowledge and extracting relevant entities and relationships are challenging and often require significant time, especially when dealing with user-generated content. Such content is typically unstructured and highly variable, making the process even more arduous. Consequently, there is a growing demand for innovative methods capable of efficiently constructing knowledge graph and extracting knowledge, even from limited or less-than-ideal sources. The development of large language models (LLMs) offering potential solutions to these challenges. LLMs possess the capability to acquire and process domain-specific knowledge, transforming unstructured data into a structured format (Bang et al. 2023; Hu, Li, et al. 2023; Qiu and Jin 2023). However, the effectiveness of outputs is notably influenced by the prompts it receives (White et al. 2023; Tian et al. 2024). The prompt-dependent nature of LLMs highlights a need for enhancing its capabilities through external knowledge sources. Given the challenges in constructing KG and the mixed benefits and limitations of LLMs, there arises a compelling case for exploring the integration of KG with LLMs. It is argued that the integration can be categorised into three approaches (Pan, Li, et al. 2024):

- KG-Enhanced LLMs: This approach involves incorporating KG during the pre-training and inference phases of LLMs to enhance their comprehension of knowledge. An application in manufacturing domain exemplifies this, where applying fine-tuning to LLMs

incorporates semantic knowledge, thereby optimising the models' responses to meet specific industry needs (Xia et al. 2024).

- LLMs-Augmented KG: LLMs are utilised for various KG tasks such as embedding, completion, construction, graph-to-text generation, and question-answering. This application is demonstrated in process KG construction and refinement for graph-based process planning, enhancing the process of knowledge completion (Xiao et al. 2023). Besides, it is argued that LLMs have the potential to assist in automatic and adaptive KG construction, addressing issues of data quality and consistency in engineering design (Pan, Luo, et al. 2024).
- Synergized LLMs + KG: This collaborative model ensures that both LLMs and KGs contribute equally and benefit mutually, facilitating bidirectional reasoning driven by a blend of data and knowledge. A notable implementation of this approach is seen in the facilitation of efficient knowledge reuse for the assembly process (Hu, Tian, et al. 2023). It supports a question-answering system that combines coarse-grained parsing of queries with fine-grained analysis using the KG.

Despite the promising applications of the integration of KG and LLMs in various domains, there remains a notable gap in the context of engineering design, particularly in the application of post-hoc explainability. As such, this paper explores using ChatGPT to assist knowledge acquisition and streamline KG construction for enhancing post-hoc explainability in design context.

3. Methodology

The process of generating explanations for ML-driven design outcomes via the integration of a KG and ChatGPT unfolds in three stages, knowledge acquisition, KG construction, and KG deployment, depicted in Figure 1.

3.1. Knowledge acquisition

In Stage I, the focus is on accumulating data, information, and knowledge relevant to ML models and design tasks. This foundational phase is crucial for generating contextual explanations and harmonising domain-specific and model-specific insights. Steps 1.1 and 1.2 outline the ML model development process, specifically adapted for design tasks. Design data can be sourced from a variety of channels, including user-generated content, third-party providers, and public databases. Preliminary operations such as data cleansing and transformation are employed to format this raw data into a structured dataset. Depending on the specific design task, the nature of data collected can span modalities like text, images, and tabulated formats. The nature of this data often dictates the choice of the most fitting ML model. The subsequent stages of model training and evaluation are driven by the objective to refine, optimise, and achieve a high-performance ML model.

In Step 1.3, a large amount of data instances sampled from training datasets is paired with the model formulated in Step 1.2 to serve as input for post-hoc XAI methods, producing localised feature-based explanations for each data instance. The selection of these XAI methods is often guided by the specificities of the model and the modalities of the input

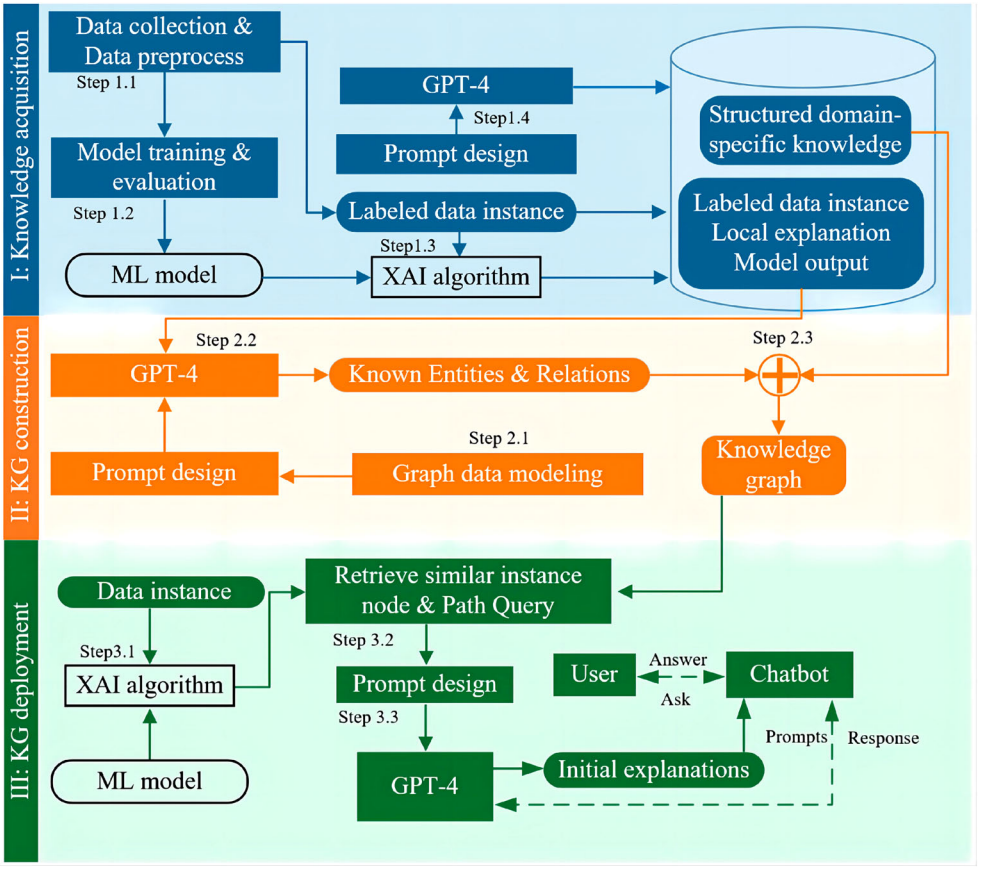


Figure 1. The process of generating contextual explanations for predictions.

data. It is imperative to weigh the merits and drawbacks of the chosen XAI methods. In this context, a myriad of XAI software solutions, including InterpretML (Nori et al. 2019), and Alibi (Klaise et al. 2021), have been devised to facilitate post-hoc explainability. A selection of these XAI methods can be found in Table 1.

In step 1.4, the objective of this step is to identify and extract structured knowledge by ChatGPT, which can be represented as the relationships between entities. Therefore, it is first essential to gather relevant entities and the attributes that are used to describe the entities. Entities consist of various kinds of features and labels. Therefore, to acquire more relevant features, ChatGPT is deployed to break down internal features into more granular sub-features or to amalgamate them features into comprehensive meta-features. This granularity allows for a more nuanced understanding of the predictive elements within the model. Additionally, ChatGPT is implemented to identify and integrate external features, which are variables outside the initial dataset. This holistic view ensures that the model captures a wider spectrum of influential factors. Once appropriate features have been selected, relations can be acquired by posting prompt sentences to ChatGPT regarding their interdependencies. To effectively acquire structured domain-specific knowledge, it is crucial to meticulously craft prompts to generate knowledge in specific templates and formats (Tian

Table 1. A brief overview of main post-hoc XAI methods

Supported methods	Data type	Advantages	Disadvantages	Explanation scope
Grad-CAM, Grad-CAM++	Text Image	(i) Applies to a broad range of CNN model families.	(i) Individual interpretations are difficult to aggregate for global knowledge.	Global & Local
Counterfactual	Tabular Text Image	(i) Provides specific and actionable explanations for individual instances; (ii) Helps identify the causal effect of input features.	(i) Computationally expensive and may not scale well with high-dimensional data	Local
LIME	Tabular Text Image	(i) Suitable to a very large number of explanatory variables, sparse explainer; (ii) Provides local fidelity; (iii) Makes no assumptions about the model.	(i) Incapable of explaining models with non-linear decision boundaries; (ii) Incapable of explaining surrounding observations.	Local
Score-CAM	Image	(i) Achieves better visualization and fair interpretation.	(i) Smoothing generates inconsistent explanations.	Local
Anchors	Tabular Text Image	(i) Localization results are poor and lead to non-interpretability; (ii) Smoothing generates inconsistent explanations.	(i) Coverage drastically decreases with an increase in the number of feature predicates	Local
SHAP	Tabular Text Image	(i) Computes many Shapley values; (ii) All Shapley values advantages connect LIME; (iii) Fast implementation for tree-based models.	(i) TreeSHAP produces unintuitive feature attributions; (ii) Does not provide causality.	Global & Local

et al. 2024). The construction of these prompts incorporates a clear background description, articulates the specific task at hand, defines any output constraints, and provides an example of the desired output format. Utilising ChatGPT in this manner enables simultaneous generation and extraction of domain-specific knowledge, including entities and their interrelations. This dual functionality facilitates the streamlining of the KG construction process, potentially reducing the steps and complexity traditionally involved in populating a KG.

3.2. Knowledge graph construction

KG can be constructed using various structures, including property graphs and Resource Description Framework graphs. This section concentrates on constructing a property graph with Neo4j, a system optimised for storing and processing graph data, commonly applied in KG, network analysis, and connected data applications (Hodler and Needham 2022). The construction begins with graph data modelling, a crucial step for establishing the

framework of the instance model. Graph data modelling necessitates a meticulous analysis of the labelled data instance, ML model output, local explanation, and domain-specific knowledge generated in Stage I. Following the acquisition of knowledge from Stage I, ChatGPT can be strategically utilised to facilitate the generation of a graph data model. This is achieved through carefully designed prompts that incorporate the acquired domain-specific knowledge, labelled data instances, detailed task descriptions, and specific output constraints. These prompts guide ChatGPT to synthesise and structure the necessary elements for the graph data model, aligning with the requirements of the property graph in Neo4j. Specifically, the graph G can be defined as a tuple $G = (V, E, A_V, A_E)$ where V is the set of nodes, E is the set of edges, A_V is the attribute function for nodes and A_E is the attribute function for edges.

3.2.1. Defining nodes and attributes

Let D represent the set of all existing data instance entities, F represents the set of all internal entities, EF represent the set of all expanded features entities, XF represent the set of all external feature entities, and L represent the set of all label entities. Thus, the set of all nodes can be represented as: $V = D \cup F \cup EF \cup XF \cup L$. The attribute function for nodes, A_V, can be described for a node $v \in V$ as:

$$\begin{cases} (f_1, f_2, \ldots, f_n, l) & \text{if } v \in D \\ (desc) & \text{if } v \in F \cup EF \cup XF \\ (label_value) & \text{if } v \in L \end{cases}$$

Here, $(f_1, f_2, \ldots, f_n)$ are the feature values for a data instance, l is the label of the data instance, and *desc* is the description of a feature.

3.2.2. Defining edges and relationships

Let R_{DF} represents the 'is_affected_by' relation between D and F. It is mentioned that only the top 20% most important internal features are selected. R_{DEF} represents the 'has_a_feature' relation between D and EF. R_{FEF} represents the 'is_subclass_of' relations between EF and F. R_C represents the 'weak_ coupling' and 'strong_coupling' relations among EF,XF, and L. The set of all edges is represented as:$E = R_{DF} \cup R_{DEF} \cup R_{FEF} \cup R_C$. The attribute for edges, A_E, can be described for an edge $e \in E$ as:

$$\begin{cases} (importance_value) & \text{if } e \in R_{DF} \\ (subclass_description) & \text{if } e \in R_{FEF} \\ (coupling_description) & \text{if } e \in R_C \end{cases}$$

Here, *importance_value* is the feature importance value for an edge between a data instance and an internal feature, *subclass_attribute* describes the subclass relationship between features, and *coupling_description* describes the reasons for coupling.

In Step 2.2 and Step 2.3, to effectively construct KG, precise extraction of entities and their interrelationships from the domain-specific knowledge is crucial. As established in Stage I, knowledge derived via ChatGPT is pre-structured, facilitating its direct incorporation into the graph data model. For the extraction of entities and relationships from labelled data instances, local explanations, and ML model outputs, prompts are methodically designed. For labelled data instances, the extraction process is straightforward: the feature values

and their corresponding labels are retrieved directly from the original dataset files. Regarding local, feature-based explanations and corresponding ML model outputs, the significant features and their importance scores are also extracted and formatted into CSV files. This structured data is then utilised to construct the instance KG within the Neo4j framework efficiently.

3.3. Knowledge graph deployment

The third stage of the proposed approach involves deploying KG in a real-world application. In step 3.1, data instances are taken from user inputs, predictions are provided using the trained ML model, and the same XAI method used in stage I is employed to provide local feature-based explanations. These explanations serve as the basis for retrieving analogous nodes in the KG. In step 3.2, both structural and content-based similarity are considered to retrieve analogous data instances in the KG. Content-based similarity measures the degree of resemblance between two data instances concerning their feature attributes. Cosine similarity, a widely employed metric for this purpose, quantifies similarity as the cosine of the angle between two feature vectors in a multidimensional space (Rahutomo, Kitasuka, and Aritsugi 2012). It is calculated by Equation (1), where $A = [f_1, f_2, \ldots, f_n, l]$ is the user input data instance and $B = [f_1, f_2, \ldots, f_n, l]$ is the feature vector of the candidate data instance node in KG. To mitigate the diminishing utility of distance measures in high-dimensional spaces for Equation (1), only the top 20% most important features are considered. Let I_A be set of the 20% most important features for data instance A, which is indicated by the local feature-based explanations for A. Then the cosine similarity considering only these features is calculated by Equation (2). In terms of structural similarity, it measures how similar two data instances are in terms of their relationships to other feature nodes in the KG (Maillot and Bobed 2018). In the KG, Neighbours (B) are the sets of internal feature nodes and expanded feature nodes connected to B. Neighbours (A) are the sets of the top 20% most important features that have influenced the ML model predictions and corresponding expanded features. As such, structural similarity can be derived by Equation (3). To combine the structural similarity and content-based similarity, these similarities are then merged using a weighted sum, as calculated by Equation (4), to derive the most similar data instance node.

$$\textbf{\textit{Cosine Similarity}}(\boldsymbol{A}, \boldsymbol{B}) = \frac{\boldsymbol{A} \cdot \boldsymbol{B}}{||\boldsymbol{AB}||} \tag{1}$$

$$\textbf{\textit{Cosine Similarity}}_{\textbf{\textit{top}}20\%}(\boldsymbol{A}, \boldsymbol{B}) = \frac{\sum_{i \in I_A} \boldsymbol{A_i} \times \boldsymbol{B_i}}{\sqrt{\sum_{i \in I_A} \boldsymbol{A_i^2}} \times \sqrt{\sum_{i \in I_A} \boldsymbol{B_i^2}}} \tag{2}$$

$$\textbf{\textit{Jaccard Coefficient}}(\boldsymbol{A}, \boldsymbol{B}) = \frac{|\textbf{\textit{Neighbors}}(\boldsymbol{A}) \cap \textbf{\textit{Neighbors}}(\boldsymbol{B})|}{|\textbf{\textit{Neighbors}}(\boldsymbol{A}) \cup \textbf{\textit{Neighbors}}(\boldsymbol{B})|} \tag{3}$$

$$\textbf{\textit{Combined Similarity}} = 0.5\ \textbf{\textit{Cosine Similarity}} + 0.5\ \textbf{\textit{Jaccard Coefficient}} \tag{4}$$

Afterwards, the algorithm commences by querying the KG to identify expanded features, utilising the input data instance A and its subset of important features I_A. It then proceeds to seek out external features that exhibit a 'strong_coupling' relationship with these expanded features. Subsequently, the focus shifts to both the expanded and external features that

Table 2. Query process for domain-specific knowledge.

Algorithm 1: Retrieve domain-specific knowledge

```
function EXTRACT_DOMAIN_KNOWLEDGE(A)
    I_A ← IDENTIFY_IMPORTANT_FEATURES(A) // Use a ML model or XAI technique
    E_A ← EXTRACT_EXPANDED_FEATURES(I_A) // Based on the value mapping of data instance features
    // Query the KG for expanded features
    E_A_NODES ← RUN_CYPHER_QUERY("
        MATCH (ef:ExpandedFeature)
        WHERE ef.name IN E_A
        RETURN ef")
    // Query the KG for external features strongly coupled with expanded features
    X_A ← RUN_CYPHER_QUERY("
        MATCH (ef:ExpandedFeature)-[:STRONG_COUPLING]- > (xf:ExternalFeature)
        WHERE ef IN E_A_NODES
        RETURN xf")
    // Query the KG for labels strongly coupled with expanded and external features
    L_A ← RUN_CYPHER_QUERY("
        MATCH (ef:ExpandedFeature)-[:STRONG_COUPLING]- > (l:Label),
        (xf:ExternalFeature)-[:STRONG_COUPLING]- > (l)
        WHERE ef IN E_A_NODES AND xf IN X_A
        RETURN l")
    // Retrieve the 'coupling_description' for the strong couplings
    C_DESC ← RUN_CYPHER_QUERY("
        MATCH (ef:ExpandedFeature)-[r:STRONG_COUPLING]- > (xf:ExternalFeature)
        WHERE ef IN E_A_NODES AND xf IN X_A
        RETURN r.coupling_description AS description
        ")
    return COMBINE_DOMAIN_KNOWLEDGE(E_A_NODES, X_A, L_A, C_DESC)
    end function
```

maintain a 'strong coupling relationship with the associated label nodes. The query also retrieves the '*coupling_description*' attribute, offering insights into the relevance of each relationship. This query process is methodically detailed in Table 2.

In Step 3.3, the knowledge extracted from the KG, comprising similar data instances with their labels, domain-specific knowledge, the input data instance, and its local feature-based explanations, is utilised to craft a prompt for ChatGPT. This prompt is carefully crafted to prompt ChatGPT into integrating this information and generating a contextual explanation. These initial explanations provide a baseline understanding for end-users. ChatGPT's interactive features then allow for an exploratory dialogue, enabling users to probe further into the explanations for enhanced clarity.

4. A case study of customer segmentation

4.1. Background

Customer segmentation forms a vital cornerstone in the product development process, involving the division of a diverse customer base into distinct segments, based on shared demographic, psychographic, behavioural, and geographic attributes (Zhang, Jiao, and Ma 2007). Working in conjunction with market targeting and positioning, customer segmentation can significantly shape the course of product development. The knowledge derived from segmentation outcomes enables manufacturers to customise product specifications, product families, and service provisions to align with the unique needs of different customer segments (Tseng, Jiao, and Wang 2010). In this context, ML plays a pivotal role in

streamlining customer segmentation. Specifically, our research employs a multilayer neural network model designed to predict whether an individual's income either exceeds or falls below $50k. The architecture of this model comprises three layers: an input layer with 14 neurons, a hidden layer with 10 neurons, and an output layer with 2 neurons. 'ReLU' (Rectified Linear Unit) was selected the as the activation function for the neurons in our network. The dataset, retrieved from the UC Irvine ML repository, comprises 48,842 customer data instances, each characterised by 14 unique features, such as age, education, relationship status, occupation, marital status, race, gender, hours worked per week, and country of residence. The task assigned to the model is to partition these customers into two income-based segments: a low-income segment (earning $50k or less) and a high-income segment.

4.2. Operation

4.2.1. Knowledge acquisition

The foundational phase of the framework is dedicated to knowledge acquisition, where domain-specific knowledge is systematically gathered, along with a set of data instances paired with their respective model predictions and local explanations. This process of knowledge acquisition is meticulously outlined as follows.

4.2.2. Acquiring local explanations for model-specific knowledge

To determine if an individual's income surpasses the threshold of $50,000, a neural network was developed with a specified set of hyperparameters. Data preprocessing was conducted to prepare the datasets, which included steps such as cleaning missing values, normalising, or standardising numerical features, encoding categorical variables, and feature selection or extraction, followed by partitioning the data into a training and testing set with an 80:20 split. The architecture comprised a single hidden layer with 10 neurons, ensuring a balance between model complexity and computational efficiency. The training was set to run for a maximum of 500 iterations, providing the model ample opportunity to converge to a solution. ReLU for activation was employed to incorporate non-linearity into the model, enhancing its ability to capture complex patterns within the data. Additionally, L2 regularisation with an alpha value of 0.001 was implemented, which discourages overfitting by penalising large weights. The optimisation of the model's weights during training was conducted using stochastic gradient descent. To evaluate the model's performance, a confusion matrix was employed. From this, the macro-F1 score was calculated, representing the mean of precision and recall scores averaged across classes, to provide a balanced measure of the model's predictive performance. This metric is particularly informative when dealing with class imbalances. The model achieved a commendable accuracy of 0.849 and a macro-F1 score of 0.794, indicating a robust predictive capability across class designations.

LIME was then utilised to create local feature-based explanations for each prediction, an example of which is depicted in Figure 2. It illustrates the prediction results about a particular data instance, which was predicted into the high-income segment (with an overall probability of 0.81). On top of the prediction result by the neural networks, the explainer further clarified the contributions of different features to this prediction result, i.e. the features of 'marital status = Married', 'relationship = Wife', and 'Education = Masters' contributed 0.23, 0.17, and 0.16 to the prediction result, respectively. To ensure a robust knowledge

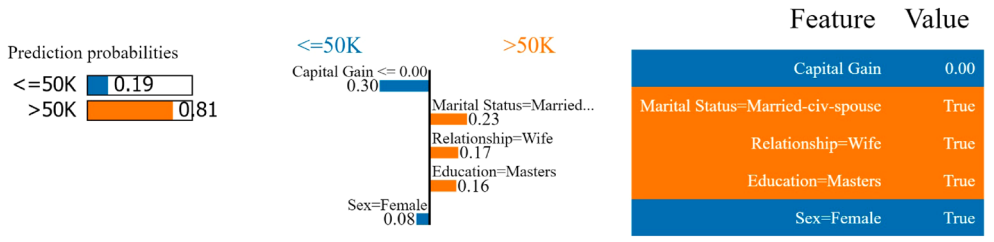


Figure 2. Feature-based explanation.

foundation for the KG, a diverse sample of 2,000 data instances was selected from the training dataset, with each instance accompanied by its local explanation generated via LIME. For the establishment of a solid knowledge foundation for the KG, a meticulous curation process was undertaken to assemble this diverse set of 2,000 data instances from the training dataset. Each instance is paired with a corresponding local explanation, produced using the LIME algorithm. Furthermore, to enhance the efficiency of entity and relation extraction in Stage II, each local explanation has been paired with its associated instance within a CSV file. This pairing facilitates a streamlined integration process.

4.2.3. Using ChatGPT to extract domain-specific knowledge

A set of critical concepts were defined to guide the extraction of domain-specific knowledge, ensuring relevance and depth in the context of income level predictions. The domain-specific knowledge pertinent to this case study primarily consists of the relationships among these features and predictions.

- Internal Features and Target: Essential 13 data variables such as age, workclass, education, occupation, relationship, marital status, race, sex, weight, capital gains/loss, hours-per-week, and income are identified for direct analysis.
- External Features: Factors external to the dataset but influential on income, such as the gender wage gap, migration trends, familial roles, and health indicators, are recognised for their contextual impact.
- Expanded Features: Discrete feature values, like age, are categorised into life stages – 'young', 'middle-aged', and 'old' – for nuanced analysis. Conversely, the synthesis of attributes into broader concepts, such as 'demographic information', leverages the composite nature of age, sex, and race.

To delineate external features, ChatGPT was employed to identify a list of approximately 40 relevant external features. The specific prompts utilised to guide ChatGPT, along with the responses it generated, are comprehensively detailed in Table 3. In the case of expanded features, ChatGPT's analytical capabilities were leveraged to dissect the dataset's 13 principal features into an extended array of 46 sub-features, enhancing the granularity of our analysis. The prompts given to ChatGPT for this detailed feature subdivision, as well as the elaborate outcomes produced, are systematically outlined in Table 4. Furthermore, ChatGPT's generative functions were harnessed to establish potential interactions and couplings between the features. Table 5 illustrates the procedure used to capture the strong

Table 3. Prompt for external features.

Prompt sentences	Output
Background: UCI census datasets are used to train machine learning models to predict whether income exceeds 50 k/yr, there are 13 internal features and 1 target. The 13 features are age, workclass, education, occupation, relationship, marital status, race, sex, weight, capital gains, and hours-per-week. Requirements: Please generate 40 external features that are related to income predictions for customer segmentation. Output constraint: the external feature should be specific and have a unique id. The output form should be like the below: feature 1, ex_id feature 2, ex_id … feature 40, ex_id For example, Vehicle Ownership, ex_1	Credit Score, ex_1 Home Ownership Status, ex_2 Mortgage Amount, ex_3 Investment Portfolio Value, ex_4 Retirement Account Balance, ex_5 Educational Attainment Level, ex_6 Number of Dependents, ex_7 Health Insurance Status, ex_8 Debt-to-Income Ratio, ex_9 Years in Current Job, ex_10 Number of Previous Employers, ex_11 Professional Certification, ex_12 … … Entertainment Expenditures, ex_38 Public Transportation Usage, ex_39 Frequency of International Travel, ex_40

coupling information for a subset of features. As such, this method is applied to deduce strong coupling information across the entire spectrum of features.

4.2.4. Knoweldge graph construction

During the knowledge acquisition phase, relevant knowledge is generated in specific templates and formats, as displayed in Tables 3–5. Each feature is uniquely identified and instantiated as an entity, assigned a distinct ID within the system. Relationships are then directly established among features, and between features and the target outcome, thus obviating the need for an additional step of relation extraction. Regarding the link between data instances and their internal features, these associations are determined based on feature importance rankings. The three most significant features, as determined by their importance scores, are connected to their corresponding data instances. The methodology for delineating these associations is detailed in Table 6, which also delineates the schema for the resulting CSV file that documents these relationships.

Utilising the CSV files, which stores entities corresponding to data instances, features, and targets, along with their interrelated associations, ChatGPT was leveraged to aid in the construction of the graph data model. ChatGPT's role involved interpreting and mapping the information from these CSV files into a structured graph data model, as illustrated in Figure 3. This process entailed ChatGPT analysing the CSV data to understand the relationships and entities, and then assisting in transforming this understanding into a coherent graph structure that accurately represents the data's interconnected nature. The resulting KG comprises 2,101 nodes and 3,255 relationships, with Table 7 showcasing a representative excerpt of these nodes and relationships.

4.2.5. Knowledge graph deployment

Data instance from the test dataset is selected and subjected to predictive analysis using a pre-established ML model. In tandem, a local explanation for the instance's prediction is generated employing an XAI algorithm, specifically LIME as used in stage I. Subsequently, several elements are aggregated: the feature vector of the instance, the machine-learned predictive classification, the top 3 ranked feature importance scores, and the expanded

Table 4. Prompt for expanded features.

Prompt sentences	Output
Background: UCI census datasets are used to train machine learning models to predict whether income exceeds 50 k/yr, there are 12 internal features and 1 target. They are age, workclass, education, occupation, relationship, marital status, race, sex, weight, capital gains, capital loss, and hours-per-week. Task requirements: please group similar values or break down single features into more detailed sub-features to capture more specific or nuanced aspects of the data. Example: the internal feature 'age' could be categorized into 'young,' 'middle-aged,' and 'old'. Output constraint: the expanded feature should be specific. The output form should be like the below: < internal_feature_1, id, new_feature_1, id, new_feature_2, id … ., mapping information between feature value and new_feature > < internal_feature_2, new_feature_1, new_feature_2 … ., mapping information between feature value and new_feature > … … . < internal_feature_13, new_feature_1, new_feature_2 … ., mapping information between feature value and new_feature > Example for output: < age, in_1, young, in_1, middle_aged, in_1_2,old, in_1_3 … , young: age < = 30, middle-aged: 30 < age < = 60, old: age > 60)	< age, in_1, young, in_1_1, middle_aged, in_1_2, old, in_1_3, young: age < = 30, middle-aged: 30 < age < = 60, old: age > 60 > < workclass, in_2, government_employees, in_2_1, private_sector_employees, in_2_2, self_employed, in_2_3, unemployed, in_2_4, government_employees: 'Federal-gov' OR 'Local-gov' OR 'State-gov', private_sector_employees: 'Private', self_employed: 'Self-emp-inc' OR 'Self-emp-not-inc', unemployed: 'Without-pay' OR 'Never-worked' > < education, in_3, primary, in_3_1, secondary, in_3_2, higher, in_3_3, primary: 'Preschool' – '5th', secondary: '6th' – '12th', higher: 'Assoc-voc' OR 'Assoc-acdm' OR 'Prof-school' OR 'Some-college' OR 'Bachelors' OR 'Masters' OR 'Doctorate' > … … < hours_per_week, in_12, part_time, in_11_1, full_time, in_11_2, over_time, in_11_3, part_time: hours_per_week < = 20, full_time: 20 < hours_per_week < = 40, over_time: hours_per_week > 40 >

features derived from the ranked attribute value mapping. These components are utilised to perform a similarity calculation with other entities within the KG. Employing combined similarity (4), the node is identified within the KG that exhibits the highest degree of similarity to the data instance. This analogous node is instrumental in providing a contextual foundation for the explanation, serving as a comparable case. Further, path queries are executed within the KG to distill domain-specific knowledge based on the algorithm in Table 2. This process furnishes the necessary contextual elements that are critical to the predictions, thereby forming the groundwork for prompt design. ChatGPT is thus engaged in articulating the explanations from multiple vantage points, enriching the context. An illustration of ChatGPT's capacity to generate contextual explanations is documented in Table 8.

Following the generation of initial explanations, the chatbot interface, powered by the contextual understanding capabilities of ChatGPT, enables end users to actively engage and seek clarifications in a manner that aligns with their individual queries, thereby facilitating an interactive explanation process.

5. Discussion and limitations

Our case study focusing on customer segmentation establishes the viability of the proposed method in generating high-fidelity explanations that are contextually rich and meaningful.

Table 5. Prompt for acquiring the coupling information among features.

Prompt sentences	Output
Background: UCI census datasets are used to train machine learning models to predict whether income exceeds 50 k/yr. there are 13 internal features and 1 target. We expand the features by subdividing the internal features, and incorporate external features. To avoid the token constraint, I will give you a part of the feature list as below: [Age, in_1: < young,in_1_1,middle-aged,in_1_2, old, in_1_3 > , Work class, in_2: < government_employer, in_2_1, private_sector_employer, in_2_2, self_employed, in_2_3, unpaid_family_worker, in_2_4 >]. Task requirements: please help me generate potential coupling information among the provided two feature lists in a way like cartesian product. Also, relation is defined by graph data modeling, the relationship type among features includes < weak_coupling, strong_coupling, is_sub_class_of > The output form should be like the below: feature_1, id, Relation, feature_2, id, Relation attribute Examples for the output Young, 1.1, In_1_1, is_sub_class, In_1, age can be divided into young, middle_aged, and old_age.	in_1_1, is_sub_class_of, in_1, Young age is a subcategory of Age. in_1_2, is_sub_class_of, in_1, Middle-aged is a subcategory of Age. in_1_3, is_sub_class_of, in_1, Old age is a subcategory of Age. in_2_1, is_sub_class_of, in_2, Government employer is a subcategory of Work class. in_2_2, is_sub_class_of, in_2, Private sector employer is a subcategory of Work class. in_2_3, is_sub_class_of, in_2, Self-employed is a subcategory of Work class. in_2_4, is_sub_class_of, in_2, Unpaid family worker is a subcategory of Work class in_1_1, weak_coupling, in_2_3, Young individuals may have a weak coupling with self-employment due to lesser experience. in_1_2, strong_coupling, in_2_2, Middle-aged individuals may have a strong coupling with the private sector as this is a common period for stable employment in one's career. in_1_3, weak_coupling, in_2_4, Older individuals may have a weak coupling with being unpaid family workers, as this is less common for this age group. in_1_2, weak_coupling, in_2_1, Middle-aged individuals may have a weak coupling with government employment, based on the assumption that government jobs might be evenly distributed among age groups. in_1_1, no_coupling, in_2_4, There is no direct coupling between young individuals and being an unpaid family worker, as these characteristics do not have a direct or clear relationship

This framework leverages a hybrid approach, which synergistically integrates the strengths of both ChatGPT and KG. As a knowledge aggregator, ChatGPT helps expand and optimise the construction of KG by generating domain-specific knowledge in specific templates and formats. By synergistically fusing model-specific and domain-specific knowledge, ChatGPT's robust reasoning prowess lays the groundwork for creating human-like explanations that enhance the explainability of ML models' output. Additionally, the robust communication abilities of ChatGPT elevate its role beyond mere explanation generation. It serves as an interactive user interface that fosters a two-way dialogue between the designers and the model. This interactive aspect allows designers to ask questions related to the explanations seamlessly, eliciting clarifications or additional information as needed. In contrast to other language models like BERT, which excels in contextual understanding within sentences, ChatGPT offers a broader range of advantages in this context (Zhou, Li, et al. 2023; Zhong et al. 2023). Its ability to generate coherent discourses, simulate interactive conversations, and provide reasoned explanations positions it as a superior choice for tasks that not only require contextual understanding but also the generation of comprehensive, user-friendly explanations.

In terms of KG, KG first serves as a structured repository for domain-specific knowledge, extracted from ChatGPT through carefully crafted prompts. The graph data model also informs prompt design, ensuring that the knowledge generated is comprehensive and

Table 6. Prompt for extracting relations for data instances and internal features.

Prompt sentences	Output
Background: A csv file is uploaded. The uploaded csv file contains 2000 data instances and aligned local explanations. The csv head and one of row is shown as follow: instance_index,predicted_value,true_label,pred_proba,top_features,Age,Workclass,Education,Marital Status,Occupation,Relationship,Race,Sex,Capital Gain,Capital Loss,Hours per week,Country 3582,low income,low income,0.9941623874913316,"[('Capital Gain > 0.00', 0.2818832792047879), ('Marital Status = Never-married', −0.16770794102061234), ('Occupation = Other-service', −0.11315234931558449)]",23,Private,565313,Some-college,Never-married,Other-service,Own-child,Black,Male,Low gain,Low loss,Long,United-States Feature entity id list: < age, in_1, workclass, in_2, education, in_3, occupation, in_4, relationship, in_5, marital_status, in_6, race, in_7, sex, in_8, weight, in_9, capital_gains, in_10, capital_loss, in_11, hours_per_week, in_12 > Task requirements: please help me extract 100 the relation between data instance and feature for the above data instance. And output the relation in csv The output form should be like the below: Data_instance_id, Relation, feature_entity_id, feature importance value Examples for the output: 4, is_affected_by,in_5, 0.249	3582, is_affected_by, in_10, 0.2818832792047879 3582, is_affected_by, in_6, −0.16770794102061234 3582, is_affected_by, in_4, −0.11315234931558449 1363, is_affected_by, in_10, −0.2607595838631323, 1363, is_affected_by, in_6, 0.23077595778198223, 1363, is_affected_by, in_8, 0.08141453422020023,

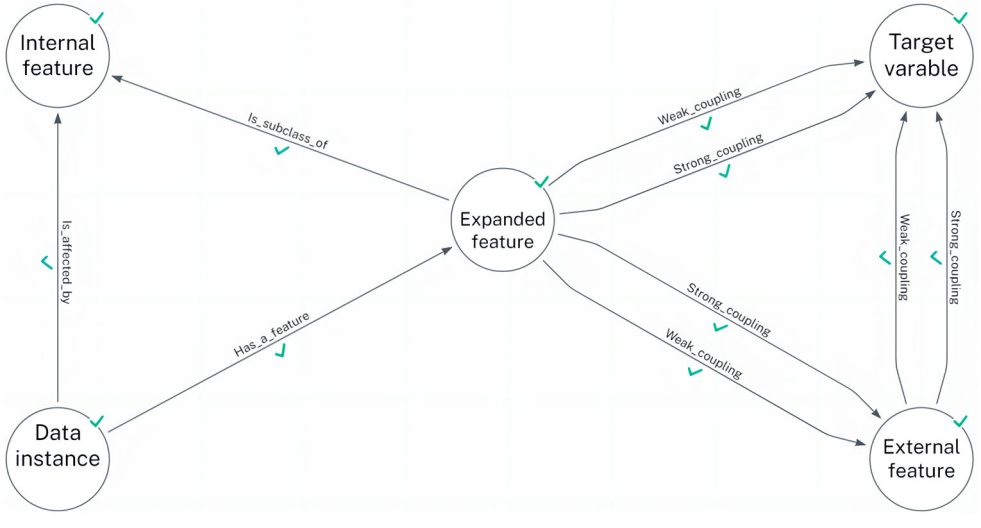


Figure 3. Neo4j graph data model.

accurate. Second, KG is used to craft contextual explanations for ML models predictions. The KG's systematic catalog of domain knowledge is leveraged to develop prompts that direct ChatGPT to generate nuanced, context-rich explanations. These prompts are rooted in the KG's mapped relationships and dependencies, ensuring deep alignment with the domain's

Table 7. A partial list of the knowledge graph.

Node			Relation		Node		
ID	Name	attribute	Name	Attribute	ID	Name	attribute
in_3_6	graduate_degree	None	Strong_coupling		t_1	low_income	None
in_1_1	young	None	Is_subclass_of	age can be divided into young	in_1	age	None
3582	data instance	Its feature vector and label: $A = [f_1, f_2, \ldots, f_n, l]$	Is_affected_by	Feature importance: 0.28	in_9	capital_gains	None
In_4_1	white_collar	None	Strong_coupling	white_collar tends to have a high_income	t_2	high_income	None
…	…	…	…	…	…	…	…
In_3_1	no_high_school	None	Weak_coupling	There is no relation between them	Ex_9	Debt-to-Income Ratio	None

ontology. Finally, KGs serve as a benchmark model to evaluate the precision and pertinence of ChatGPT's explanations. The KG's interconnected nodes and relationships form a robust framework that connects AI-generated narratives to empirical data points. This integration ensures that explanations are anchored to the data's foundational structure, maintaining the integrity and reliability of information conveyed to users.

In our approach to XAI, LIME is employed for extracting detailed insights from black-box models at an instance level. This choice is pivotal for ensuring that our explanations are not just comprehensible but also precisely aligned with the model's specific decision-making processes for each instance. Such alignment is essential to maintain the high fidelity of our explanations, guaranteeing that they accurately reflect the model's internal workings. LIME also acts as a vital bridge linking AI decision-making with our KG. By pinpointing the most influential features in the model's predictions, LIME informs the development and refinement of the KG. This process involves defining entities and relationships within the KG that mirror the influential data instances and features identified by LIME. This integration ensures that the KG serves not only as a repository of domain knowledge but is also intricately connected to the operational logic of the ML model.

ChatGPT assumes a pivotal role in the intricate processes of knowledge acquisition, KG construction, and KG deployment. Central to harnessing the full potential of ChatGPT is the art and science of prompt engineering, which profoundly influences the efficacy of each stage. During the knowledge acquisition phase, the design of prompts is particularly critical. It requires a nuanced approach to elicit domain-specific knowledge that is not only accurate but also appropriately structured for seamless integration into the KG. The quality of these prompts directly correlates with the integrity and utility of the graph data model, as they determine the structure and granularity of the curated information. As the transition to KG construction takes place, the precision of the prompts becomes even more consequential. Effective prompt design enables the direct elicitation of knowledge, obviating the need for additional layers of entity and relation extraction. Furthermore, in the deployment phase, the role of prompt engineering extends to the synthesis of domain-specific and model-specific knowledge. Here, the prompts act as a bridge, fusing disparate pieces of information to generate contextual explanations. These explanations are not only integral to interpreting the model's output but also to providing end-users with understandable insights. The adaptability of prompt design allows for exploration beyond the initial scope. By refining prompts based on the feedback loop of domain-specific and model-specific

Table 8. Prompt for generating contextual explanations.

Prompt sentences	Output
Background: a machine learning model is used to predict whether the input data instance is over 50k/year. Input data instance: [Age:54, Workclass: Private, Education: HS-grad, Marital Status: Married-civ-spouse, Occupation: Exec-managerial, Relations: husband, Race: White, Capital Gain: 4185, Capital Loss: 0, Hours per week: 35, Country: Unite-States] Model prediction: High income Top three feature importance (enabled by LIME explainer): [('Capital Gain = 4185', 0.2956996448551376), ('Marital Status = Married-civ-spouse', 0.231669349203099), ('Sex = Male', 0.07140066505809566)] The content retrieved from KG includi ng the node of data instance that is like input data instance, and domain specific knowledge. Similar data instance: [Id: 7382, Age:57, Workclass: Private, Education: Bachelors, Marital Status: Married-civ-spouse, Occupation: Sales, Relations: husband, Race: White, Capital Gain: 2174, Capital Loss: 0, Hours per week: 40, Country: Unite-States] Label: High income Domain specific knowledge: 1. High_gains is strong coupling with high income feature for High-income individuals are more likely to have disposable income that they can invest in assets like stocks, real estate, or businesses that can appreciate in value over time 2. Married is strong coupling with high income feature for Married couples often have dual incomes, which naturally increases the household income. When both partners are working, the combined income is typically higher than that of a single individual, which can skew the dataset towards higher incomes for married individuals. 3. High_gains is strong coupling with graduate degree for Individuals with graduate degrees generally have access to higher-paying jobs, which can result in both higher income and higher capital gains Task requirements: please explain the prediction for the input data instance by synthesise the similar data instance node with its label, the domain specific knowledge, and feature importance generated by local explainer LIME.	The prediction for the input data instance being categorised as 'High income' can be explained by synthesising the information from the similar data instance, the domain-specific knowledge, and the feature importance as provided by the LIME explainer. Firstly, looking at the top three feature importances provided by the LIME explainer, we can see that a 'Capital Gain = 4185' has the highest influence on the prediction. This aligns with the domain-specific knowledge stating that 'High_gains' is strongly coupled with a high-income feature. It's likely that individuals with capital gains have additional sources of wealth that contribute to a higher income level, which is consistent with the information that high-income individuals often have investments that appreciate over time, leading to capital gains. Secondly, the 'Marital Status = Married-civ-spouse' feature also has a significant impact on the prediction. This is corroborated by the domain knowledge that married couples often benefit from dual incomes, which can result in a higher combined household income. Since both the input data instance and the similar data instance have marital statuses listed as married to a civilian spouse, this attribute likely reinforces the model's prediction of a high income. Thirdly, the feature 'Sex = Male' has some influence on the prediction. While it's not specified in the input data instance, we can infer from the 'Relations: husband' attribute that the individual is male. This feature's impact is less pronounced compared to the other two but still contributes to the prediction outcome. When comparing the input data instance to a similar data instance with a label of 'High income', we can observe some commonalities. Both instances have individuals who are in their mid-late 50s, work in the private sector, are married to a civilian spouse, and have capital gains. However, the similar data instance has a higher level of education (Bachelors vs. HS-grad) and works more hours per week (40 vs. 35), which could also be factors contributing to high income, as noted in the domain-specific knowledge which associates graduate degrees with higher-paying jobs and thus higher income and capital gains. By synthesising these insights, we can conclude that the model's prediction of 'High income' for the input data instance is largely due to the significant capital gains, the status of being married to a civilian spouse, and being male, all of which are associated with higher income levels according to the provided domain-specific knowledge and empirical data.

knowledge, ChatGPT can be tailored to pursue various analytical objectives and unearth a broader spectrum of insights. This adaptive approach highlights the potential of prompt engineering to pivot in the direction of analysis, unveiling nuanced dimensions of the data.

It is important to recognise and address its limitations. First, the challenge of feature modality remains significant. Critical features identified through XAI often exist in a variety of modalities, such as textual, numeric, and imagery. This diversity introduces an additional layer of complexity when querying the relevant domain knowledge. As it currently stands, our KG structure is not equipped to fully support multi-modal features. Consequently, the querying of knowledge using non-textual, image-based features remains a hurdle. Developing mechanisms to handle multi-modal data, particularly in the context of querying and forming connections within the KG, is therefore a critical area for future refinement. Second, the construction of our KG hinges upon the ability of ChatGPT to generate knowledge based on well-crafted prompts. However, the quality of the knowledge generated may significantly vary, contingent on the design of these prompts. This variability exposes us to the risk of incorporating inaccurate or less reliable knowledge into the KG. The authenticity and relevance of knowledge generated by ChatGPT should be further validated by designers through human machine interactions. Last, the efficacy of the contextual explanations and the interactive process developed to augment human-machine collaboration has not been empirically tested with human subjects within this study.

6. Conclusion

In conclusion, this paper presents an innovative methodology for enhancing post-hoc explainability in ML-driven design by leveraging ChatGPT and KG. Our method is aimed at generating more meaningful, insightful, and trustworthy explanations to aid designers in making informed decisions, enhancing their confidence in discerning the trustworthiness of AI outputs. From our theoretical investigation and practical case study of customer segmentation, two key research findings have emerged regarding the hybrid approach to explainable ML-driven design. First, compared to traditional KG construction, where domain knowledge can be challenging to retrieve and collect, ChatGPT is utilised primarily to widen the avenues of knowledge sourcing. This resource-expanding role of ChatGPT can potentially transform the construction and refinement of KG. Second, compared to traditional XAI methods, our proposed approach automates the generation of more contextually rich explanations and enables an interactive explanation process, which align more effectively with the explanation requirements of designers and improve post-hoc explainability within ML-driven design. Furthermore, considering the rapid development of ChatGPT, particularly its multimodal capabilities, significant implications for the future of KG are anticipated. The integration of ChatGPT's advanced processing abilities with KG can lead to more dynamic and comprehensive knowledge systems. These systems are expected to be capable of not only understanding and processing textual information but also interpreting visual and auditory data, thereby enriching the knowledge base and context available within KG. This evolution can fundamentally alter how KG is constructed, updated, and utilised, making them more versatile and robust in handling complex, multimodal data environments. Through these findings, our research contributes to the literature on XAI and ML-driven design and underscores the potential of LLMs like ChatGPT and KG in enhancing the post-hoc explainability of ML-driven decisions. Future research will continue to explore

and address the limitations identified in this study, with a keen focus on harnessing the rapid advancements in LLMs to further enhance the efficacy and practicality of KG.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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